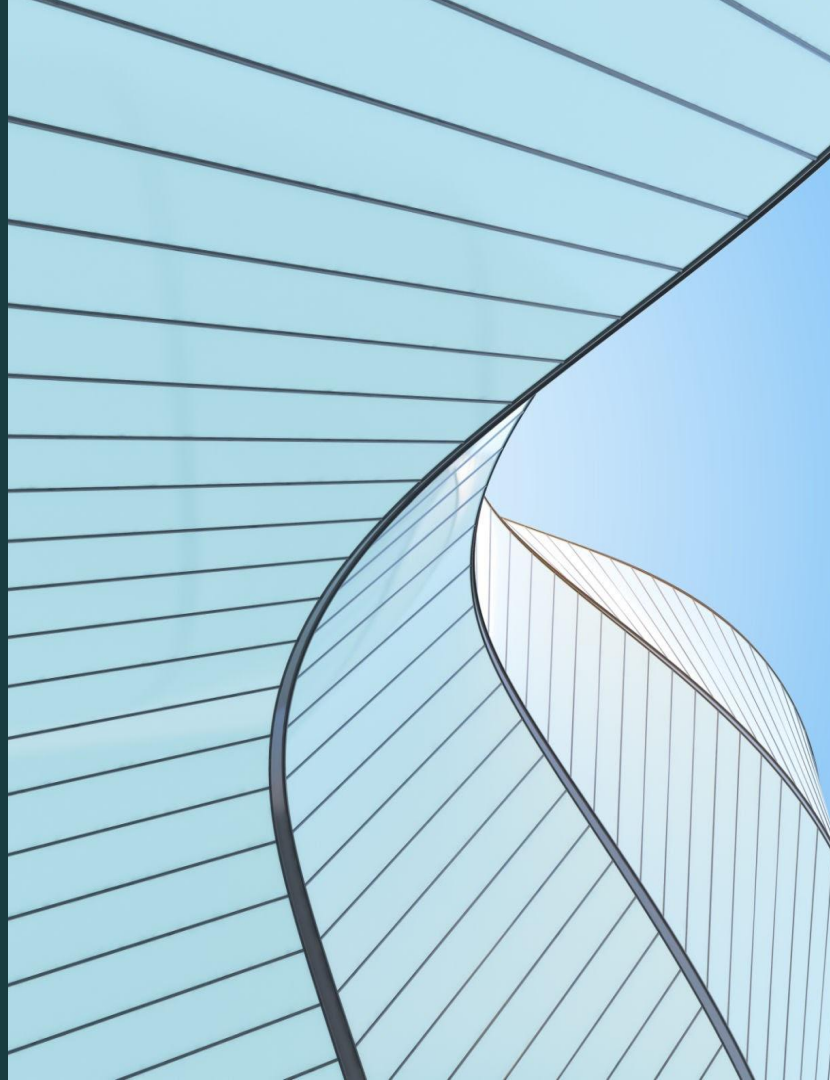
Grid Stress Monitoring, Demand, & Growth Modeling

DS591 Team 08 Project Presentation

Members: Alexander Shah, Aijia Zhang, Samantha Gibson, Jiaxi Li

[Our team does not consent to video or audio footage of ourselves to be fed into any generative AI models]



Problem: Electricity usage is on the rise

Industrial growth & Compute intensive infrastructure stressing existing grids



SOLUTION

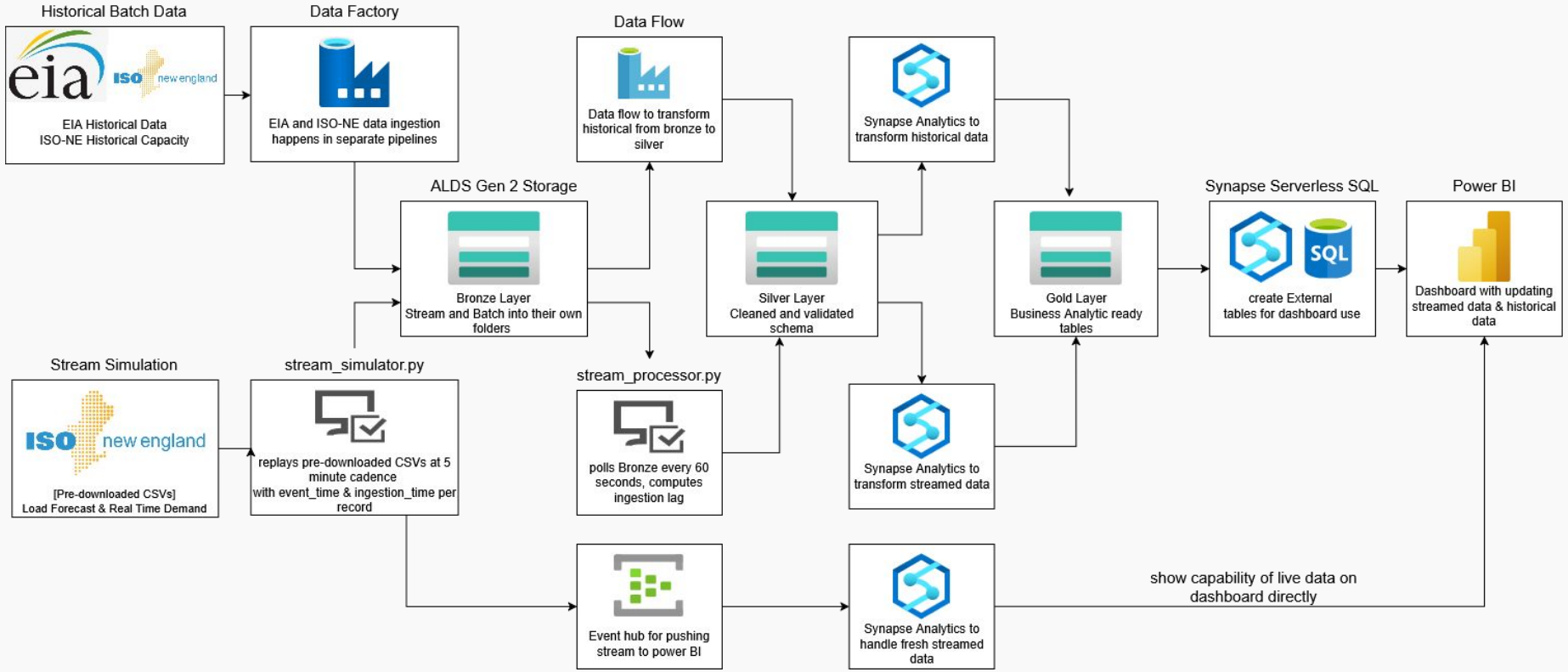
A pipeline that combines real-time grid stress & long-term demand growth

- Streaming data: ISO-New England
  - System demand every five minutes
  - Five-minutes load forecast
- Historical Data: EIA (Energy Information Administration)
  - Historical electricity consumption by region
  - Supplementing with ISO-NE Historical data

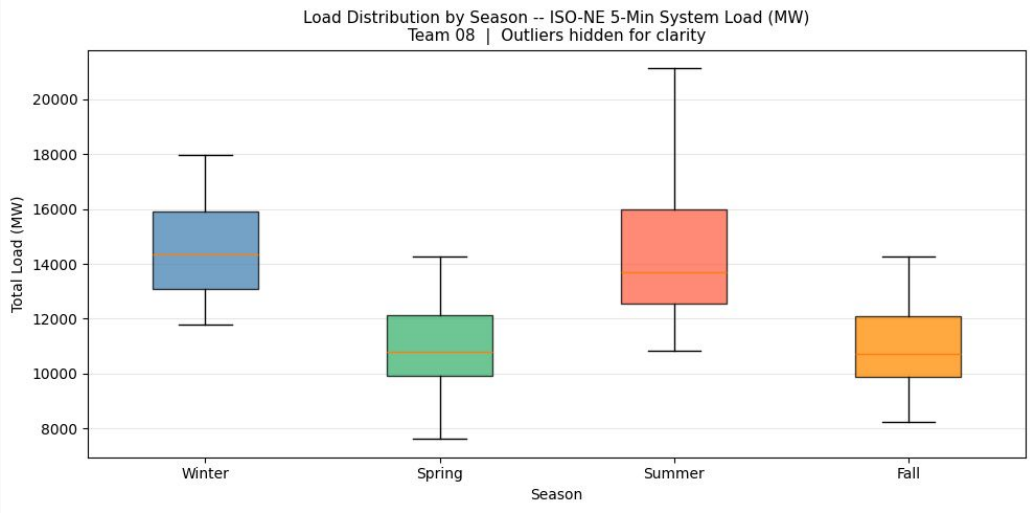
Our *Initial Predictions...*

1) Seasonal Demand Patterns
2) New AI Infrastructure Impact
3) Weekly Operation Patterns
4) Forecast Accuracy Degradation

ARCHITECTURE



EDA



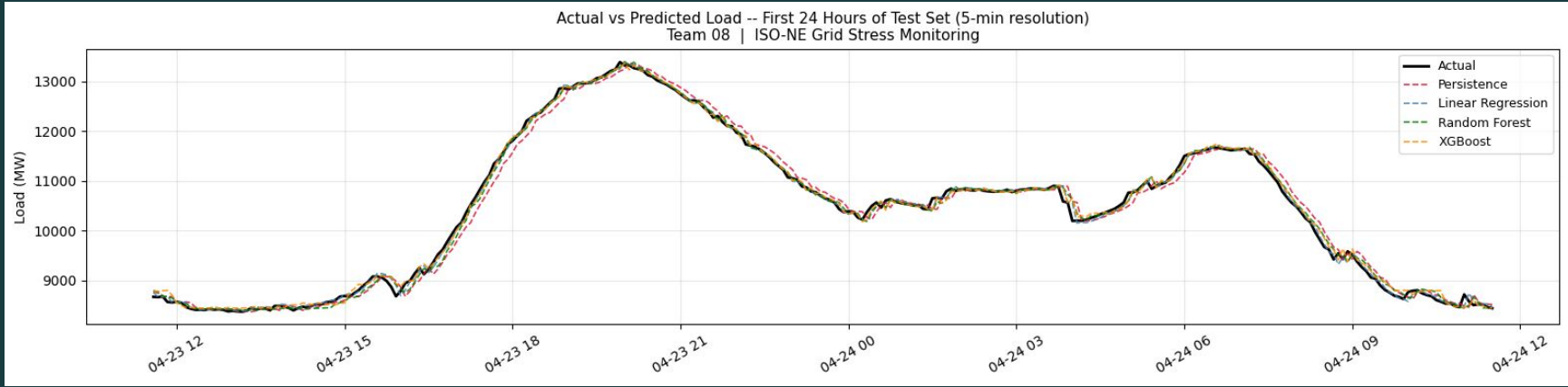
Batch Data (EIA monthly, 2016–2025):

- Residential and commercial oscillate 3,000–5,000M kWh seasonally
- Industrial stays flat and visibly declining in 10-year trend
- Geographic split: RI/MA summer peaked, ME/VT winter peaked

Streaming Data (ISO-NE 5-min, Jul 2025–May 2026):

- Load range: 7,624 MW (winter night) to 22,068 MW (summer peak)
- 439 stress events detected (4.2% of all intervals)
- Morning dip ~8,500 MW, evening spike ~12,500 MW on typical day
- Rolling 12-avg smooths volatility, hovering 10,400–10,600 MW

Model *Results*



Model	Test MAPE	RMSE	Stress F1	Overfit
Persistence	0.987%	133 MW	0.679	OK
Lin. Reg.	**0.424%**	**65 MW**	**0.829**	**OK**
Ran. Forest	0.726%	109 MW	0.826	OVERFIT!
XGBoost	0.854%	137 MW	0.746	OVERFIT!

Predictions vs *reality*

Prediction	Result	Key Evidence
1) Seasonal Patterns	Confirmed (with nuance)	RI/MA summer-peaked (1.24×), ME/VT winter-peaked (0.87×)
2) AI Infrastructure Growth	Refuted	All 6 NE states negative industrial CAGR (-0.28% to -5.68%)
3) Weekly Operational Patterns	Partial	Monthly cycles confirmed (1.53× Jul vs May), daily data needed for full test
4) Forecast Accuracy over time	Confirmed	RF F1=0.826, LR perfect recall, 19 stress episodes detected

Conclusions...

- Predictions had mixed results
- What would we do differently
- Future work

Thank you!

Questions?

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